An odd-total lower bound for simple SOC representations

Research proof and Lean formalization
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Abstract

We answer the second open question in Section 7 of Wang's 2024 paper on weighted geometric means: if the positive primitive integer weights have odd sum, every simple second-order cone representation has at least as many constraints as original input variables. The proof treats the literal existential representation, allowing repeated indices, self-loops, unused variables, and cycles initially. Equality in the previous counting bound forces an injective input map. Logarithmic balance equations and a finite doubling argument then force the total weight to be even. The full representation theorem is formally verified in Lean.

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1 The source question

Let $s_1, \dots, s_m$ be positive integers and $S = \sum_j s_j$. Consider

$$t^S \le \prod_{j=1}^{m} x_j^{s_j}, \qquad x_j, t \ge 0. \tag{1}$$

Following Wang's Definition 3.1 [1], a simple SOC representation of size $n \ge 1$ has n internal variables $y_1 = t, y_2, \dots, y_n$, the last $n-1$ being existential nonnegative auxiliaries. For each i it imposes one inequality

$$y_i^2 \le v_{c(i,0)} v_{c(i,1)}, \tag{2}$$

where v lists all $m+n$ terminal and internal variables and the input map c is arbitrary. Its projection must be exactly (1). Write $L(s_1, \dots, s_m)$ for the minimum size. Wang proved $L \ge m-1$ (Theorem 3.6) and asked on p. 1512 whether odd S, for primitive weights, always gives $L \ge m$.

Theorem 1. *For positive integers $s_1, \dots, s_m$ with odd sum, every simple SOC representation of (1) has size $n \ge m$.*

No gcd assumption is needed. In particular this proves the requested statement when $\gcd(s_1, \dots, s_m) = 1$.

We use only soundness: any nonnegative assignment satisfying all constraints must satisfy (1). The already resolved bivariate minimum-cardinality conjecture (Wang's Corollary 4.5) is a different question and is not a new result here.

2 Counting the input slots

Draw an edge from each internal vertex to each of its two child slots in (2). Keep both slots even when their input indices coincide, and let r be the output vertex. Every terminal is reachable from r. Indeed, otherwise assign value 1 to all reachable vertices and 0 to all others. Each reachable internal vertex has both children reachable, so its constraint holds with equality. Each unreachable internal vertex has $y_i^2 = 0$ and a nonnegative product on the other side. Thus every constraint holds, while $t = 1$ and a positive-weight input is zero, contradicting soundness.

If R internal vertices are reachable, all m terminals are reachable, and each reachable vertex except r is hit by a reachable child slot. Consequently

$$m + R - 1 \le 2R, \qquad m - 1 \le R \le n.$$

This reproves the old counting bound without irredundancy assumptions. Suppose for contradiction that $n < m$. Necessarily $n = R = m-1$. The map from all $2n$ child slots plus one extra slot for r onto all $m+n = 2n+1$ vertices is now a bijection. Hence the child map is injective, every nonroot vertex is used exactly once, and r is never an input.

3 Balance and the parity obstruction

Let A_{ij} count the child slots of internal vertex i equal to internal vertex j, and let B_{ij} count its terminal children equal to terminal j. Set $M = 2I - A$. Every column of A has sum at most 1. The matrix M is invertible: if $M^T w = 0$, then each $2w_j$ is either zero or a single entry of w. Taking a largest absolute entry gives $2 \max_j |w_j| \le \max_j |w_j|$, so $w = 0$.

For any $u \in \mathbb{R}^m$, solve $Mz = Bu$. The assignment of terminal values e^{u_j} and internal values e^{z_i} satisfies every constraint with equality. Soundness and strict monotonicity of the exponential give

$$Sz_r \le \sum_j s_j u_j.$$

The potential $(-u, -z)$ is also a solution, so equality holds for all u. Define the real flow vector q by

$$M^T q = Se_r.$$

Since $Mz = Bu$, we have $q^T Bu = Sz_r = s^T u$ for every u, whence $q^T B = s^T$. The unique-input conclusions identify these equations as

$$q_i = s_j \qquad \text{if a child of } i \text{ is terminal } j, \tag{3}$$

$$q_i = 2q_j \qquad \text{if a child of } i \text{ is internal } j, \tag{4}$$

$$2q_r = S. \tag{5}$$

In (4), $j \ne r$ because the root is not an input. Positivity of the flow is neither assumed nor needed.

Let $D = \{i : q_i \notin \mathbb{Z}\}$. A child of a vertex in D cannot be terminal, by (3). If it is internal j, then (4) implies $j \in D$, since an integer q_j would make q_i an integer. The injective child map therefore restricts to an injection

$$D \times \{0, 1\} \longrightarrow D.$$

Finiteness yields $2|D| \le |D|$, so D is empty. In particular q_r is an integer. Equation (5) makes S even, contrary to the hypothesis. This proves Theorem 1.

4 Formal certificate and scope

The certificate contains 41 declarations in seven modules, checked using Lean 4.34.1 and mathlib revision `d13f23b7`. Its endpoint is

`WGMParity.source_odd_total_bound`.

The formal representation is the existential projection of the literal nonnegative-real quadratic inequalities. Arbitrary finite index maps are allowed; injectivity, root exclusion, matrix invertibility, potential existence, and flow integrality are all proved. The stronger soundness-only theorem is also certified. A fresh isolated replay uses warnings as errors, has no proof holes or additional axioms, and reports only the standard axioms `propext`, `Classical.choice`, and `Quot.sound`. An independent exact enumeration of all 22,183 unordered input configurations for $2 \le m \le 4$, $n = m - 1$, including repeated inputs and cycles initially, agrees with the theorem; this finite check is not used in its proof.

The original $m - 1$ bound and the general correspondence with mediated sets are credited to Wang. The later graph-based optimization work [2, 3] supplies broader modeling and computational frameworks; our proof works directly with the existential inequalities. The documented prior-work screen is bounded, and no absolute priority claim is made. Independent review and final packet acceptance remain pending for this draft.

References

[1] J. Wang, *Weighted geometric mean, minimum mediated set, and optimal simple second-order cone representation*, SIAM J. Optim. **34** (2024), 1490–1514. doi:10.1137/22M1531257.

[2] V. Blanco and M. Martínez-Antón, *On minimal extended representations of generalized power cones*, SIAM J. Optim. **34** (2024), 3088–3111. doi:10.1137/23M1617205.

[3] V. Blanco and M. Martínez-Antón, *On the structure and computation of extremal mediated graphs with applications to conic optimization*, J. Optim. Theory Appl. **210** (2026), article 24. doi:10.1007/s10957-026-03048-5.

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